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STATE OF COLORADO

DIVISION OF MINERALS AND GEOLOGY

Department of Natural Resources

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Bill Owens
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Greg E. Walcher
Executive Director

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Division Director

September 7, 1999

Mr. John Lohr
Southdown Co., Inc.
P.O. Box 529
Lyons, CO 80540

RE: Report on Storage of Cement Kiln Dust at the Southdown Inc. Lyons Plant, Lyons Quarry,
Permit No. #M-77-208

Dear Mr. Lohr,

This letter is being written to address technical issues which need clarification in the recent report submitted by Banks & Gesso on behalf of Southdown Co., Inc., addressing the storage of Cement Kiln Dust (CKD) in C-Pit at the Lyons Quarry, Permit No. #M-77-208. Under the current permit, the Operator is allowed to store CKD in this pit. After a consultation with the Division of Minerals and Geology (DMG) in January 1999, the Operator agreed to do chemical leach tests on the CKD and submit data and hydrological modeling of the pit, so that DMG could evaluate the potential impacts of CKD storage on the local groundwater. DMG has been responsible for evaluating the impacts of mining activities on groundwater quality since 1992, and it was believed that a review of the current permit conditions was necessary in order to confirm that all concerns were being addressed.

Having received the requested report from Banks & Gesso on June 23, DMG personnel have reviewed the document and have a few concerns which still need to be addressed. Attached is the review of the report by Harry Posey. Please address the following questions:

- 1) In section 5.0, Geologic and Hydrogeologic Setting of the Disposal Area, reference is made to the various shale and limestone layers underlying and abutting C-Pit. Please clarify whether the source of this information is the NRCS soil survey, and whether the distances which these formations extend to the alluvium have been taken from this report or from field measurements.
- 2) In section 7.2, Simulation of the Potential Quantity of Leachate, it is stated that two models of the site conditions were simulated, both of which assumed a closed disposal site. Since the site will remain open for several years while CKD is still being deposited in the pit, DMG

feels it is necessary for a simulation of the leachate flux rate to be done under these conditions. DMG would also like to know which exact "worst case" quantity of leachate resulted in modeled exceedance of the water quality parameters in the alluvium, and to what extent the standards were exceeded.

- 3) In Appendix D, in the HELP data sets, a key is needed to clarify which materials are being evaluated in each layer, since they are only identified here by Material Texture Number. The basis upon which the values for layer thickness were chosen should also be made clear, since some of these numbers do not match previously mentioned distances or sedimentary layer depths.

Please submit this information as soon as possible so that DMG may complete our evaluation of the CKD disposal on site, and determine whether any modifications to your permit may become necessary. DMG hopes to make a determination regarding this portion of your permit by September 30, 1999.

If you have any further questions, please contact me at (303) 866-3567.

Sincerely,



Christina L. Kamnikar
Environmental Protection Specialist

Cc: Carl B. Mount, DMG
H. Bruce Humphries, DMG
Harry H. Posey, DMG
Paul Banks, Banks & Gesso
St. Vrain Community Watch Dogs Group

Attachment (1)